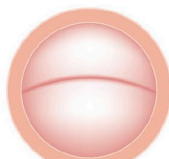
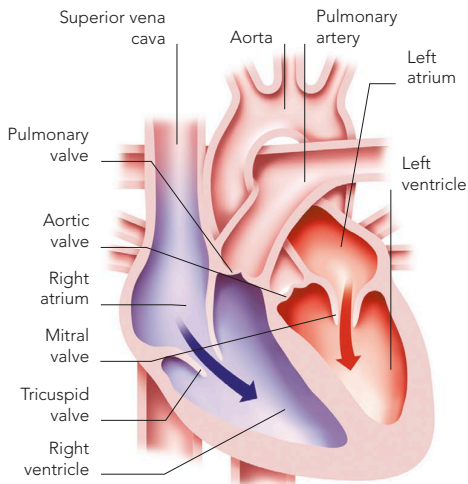


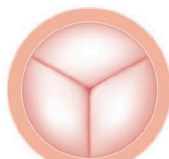
HEART VALVES

The heart has four valves to control blood flow. Each has the same basic structure, although they differ in certain details. The two atrioventricular valves lie between the atria and ventricles. The mitral valve, on the left side, has two cusps, while its right

counterpart, the tricuspid valve, has three. The two semilunar valves are at the exits from the ventricles: the pulmonary valve between the right ventricle and the pulmonary artery, and the aortic valve between the left ventricle and the aorta.



TWO CUSPS



THREE CUSPS



PULMONARY VALVE

This valve lies between the right ventricle of the heart and the pulmonary artery. It opens as the right ventricle contracts and forces blood out of the heart towards the lungs.

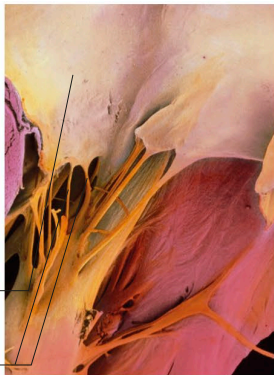
FUNCTION OF HEART VALVES

The tricuspid valve controls blood flow from the right atrium to the right ventricle; the pulmonary valve, from the right ventricle into the pulmonary artery; the mitral valve, from the left atrium to the left ventricle; and the aortic valve, from the left ventricle into the aorta.

MITRAL VALVE

This image of a healthy human heart valve shows the heart strings (chordae tendinae) and valve cusps. The mitral valve lies between the left atrium and the left ventricle.

Cusp
Chordae tendinae



Direction of blood flow
Valve cusp open

Blood pushes against valve

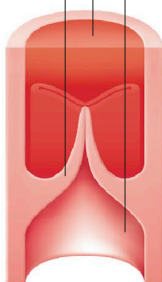
Blood at high pressure
Valve cusp shut

Blood at low pressure



HEART VALVE OPEN

The flexible cusps are forced apart by the pressure of blood as the heart contracts.



HEART VALVE CLOSED

Back pressure causes the cusps to close and seal at their edges, to stop reverse blood flow.